

An agent is a capsule shaped entity that can navigate within scenes and interact with objects.

A scene within AI2-THOR represents a virtual room that an agent can navigate and interact in. There are 4 scene Categories, each with 30 unique scenes within them: Kitchen, Living Room, Bedroom, Bathroom.

An action is a discrete command for the agent to perform within a scene (e.g. MoveAhead, RotateRight, PickupObject).

A sim object is an object that can be interacted with by the agent. Objects have a range of interactions based on their assigned `ObjectType`.

An object is considered **visible** when it satisfies two conditions:

- ## Warning

An object rendered in an image will not always be visible to the agent. For example, an object outside the 1.5 meter threshold could be seen in an image, but will be reported as not-visible to the agent.

Each object's visibility is reported **in the metadata**.

An object is said to be interactable if it is both `visible` and if it is unobstructed by any other objects. For instance, if a `Sponge` is behind `ShowerGlass`, then it will report `visible` as `True`, but the agent will be unable to interact with it due to the obstruction.

## Warning

If an object is interactable, then that implies the object is visible. But, if an object is visible, that does not imply it is interactable.

A `Receptacle` is a type of object that may be the parent of another object. Examples include `TableTop`, `Cup`, `Sofa`, `Bed`, `Desk`, and `Bowl`. The full set of `Receptacle` object types can be filtered on the [Object Types](#).

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